B.Sc. Engg. CSE 5^{th} Semester 13 April 2026 (Morning)

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

SEMESTER FINAL EXAMINATION **WINTER SEMESTER, 2024–2025**
DURATION: 3 HOURS **FULL MARKS: 150**

CSE 4513: Software Engineering and Object-Oriented Design

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 6 (six)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

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1. a) Assume you have ordered a perfume, *Versace Eros*, from an online store named *Ivaly*, and the package has just been delivered to you. Using this situation as an analogy, describe the concept of **Smoke Testing** in software development. 5 (CO3) (PO1)

   b) Consider the code shown in Code Snippet 1 with the test input `x = 11, y = -1`. Determine the percentage of **Statement Coverage** achieved by this test case. 6 (CO3) (PO1)

```
int evaluate(int x, int y) {
    int sum = x + y;
    int bonus = 0;
    int extra = 0;

    if (sum >= 10) {
        bonus = 3;
        sum = sum + bonus;
    }

    if (x > 0) {
        extra = 2;
        sum = sum + extra;
    }

    sum = sum * 2;
    return sum;
}
```

Code Snippet 1: Code snippet for Question 1.b

   c) "In software design, it is desirable to maintain strong coupling and weak cohesion." Do you agree with this statement? Provide a brief justification. 4 (CO2) (PO1)

   d) A system takes a students exam score as input within the interval $[0, 200]$. The student is considered to have passed if the score is ≥ 120; otherwise, the result is fail. Using the *robustness-based Boundary Value Analysis (BVA)* technique, construct suitable test cases for this system. Present your answer in a table including the following columns: `Test Case No., Input Value, Expected Result, and Justification.` 10 (CO3) (PO1)

2. a) Write the differences between the following pairs of concepts. Each answer should include a brief explanation of both terms and one relevant example. 2 × 5 (CO3) (PO1)

   i. Backward compatibility testing vs Forward compatibility testing

   ii. Upward scalability testing vs Downward scalability testing

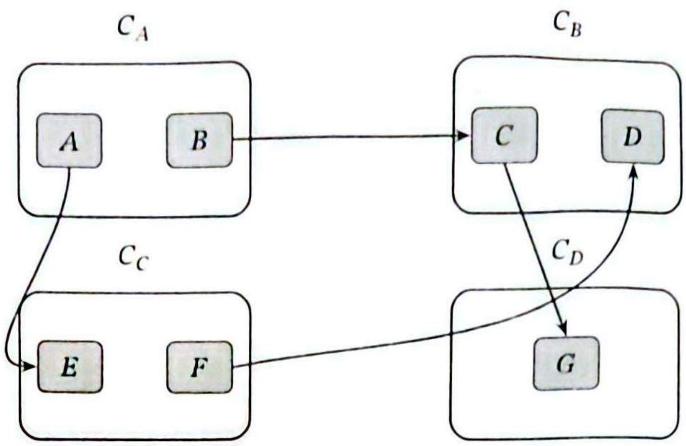

Figure 1: Component dependency diagram for Question 2.b

b) Consider the component dependency diagram shown in Figure 1. Determine the *instability* I of component C_B, showing all necessary steps in your calculation. 7 (CO2) (PO1)

c) As a software developer of *Station 23*, you are responsible for building a mobile app. While testing it, you encounter issues such as failures, bugs, and errors. Explain the meaning of *Failure*, *Error*, and *Defect/Bug*, and show the correct sequence in which they occur during software development. 8 (CO3) (PO1)

3. a) Consider a software system characterized by the following parameters: 8 (CO2) (PO1)

$$EI = 55, \quad EO = 30, \quad EQ = 50, \quad ILF = 09, \quad EIF = 04$$

Assume that all weighting factors are *High*, except *ILF*, which has *Low* complexity. In addition, the system requires *incidental performance*, *essential end-user efficiency*, *average distributed data processing*, *moderate data communication*, and *significant transaction rates*. All other General System Characteristics (GSCs) have no influence. Compute the *Function Points (FP)* of the system using *Function Point Analysis*. Use Table 1 and Table 2 where necessary. Use the constant `0.065` for the value adjustment calculation.

Table 1: Weighting factors for Function Point Analysis for Question 3.a

Measurement Parameter	Low	Average	High
External Inputs (EI)	1	2	3
External Outputs (EO)	4	5	7
External Inquiries (EQ)	8	9	16
Internal Logical Files (ILF)	7	10	15
External Interface Files (EIF)	5	8	10

Table 2: General System Characteristics (GSC) Influence Scale for Question 3.a

Value	Influence Level
0	No Influence
1	Incidental
2	Moderate
3	Average
4	Significant
5	Essential

b) A software company is building a notification management system that handles different types of notifications, including `EmailNotification`, `SMSNotification`, and `PushNotification`. The main application requests a notification object from a central component without knowing which specific notification class will be created. Based on the user's selection, the system decides which type of notification object to instantiate at runtime. Answer the following questions in the context of this scenario: $5+5+4+4$ (CO2) (PO1)

i. Identify the most suitable design pattern that should be used in this scenario and justify your answer.

ii. Later, the developers observe that creating a new notification object is computationally expensive because each notification requires loading configuration data and templates. To improve performance, the system should create a notification once and then duplicate the existing notification object whenever another similar notification is needed. Which design pattern would now be more appropriate for this requirement? Justify your answer.

iii. Suppose the developers design the class `EmailNotification` such that it inherits behaviors from two parent classes: `Notification` and `MessageService`. What object-oriented concept does this represent? Describe the concept.

iv. Assume the class `Notification` defines a method `send()`, and the class `PushNotification` provides its own implementation of the same method. What object-oriented concept is demonstrated in this situation? Explain how it works in this context.

4. a) Beta testing is commonly explained using the concept of the **3R's**. Briefly describe this concept in the context of beta testing. 6 (CO3) (PO1)

b) A software component accepts only one input variable. A tester prepares exactly five test cases using boundary-focused testing. Identify the possible Boundary Value Analysis (BVA) technique applied in this case, and briefly explain the reasoning behind your answer. 6 (CO3) (PO1)

c) Refer to the code in Code Snippet 2 and answer the following: $6+5+4$ (CO3) (PO1)

i. Construct the Control Flow Graph (CFG) for the program.

ii. Compute the cyclomatic complexity of the program using the CFG,.

iii. Identify the independent paths that should be tested based on the CFG.

```
start
if (A) then
    perform P
    if (B) then
        perform Q
    else if (C) then
        perform R
    else
        perform S
    endif
endif
end
```

Code Snippet 2: Code snippet for Question 4.c

5. a) "Regression testing is a subset of sanity testing." Do you agree with this statement? Briefly explain your answer. 5 (CO3) (PO1)

b) Why is the Big Bang approach viewed as an ineffective strategy in integration testing? Name the other two decomposition-based integration testing approaches and briefly explain why they are better. 6 + 4 (CO1) (PO1)

c) For maintaining the OBE framework of the university, IUT is developing a software system called the **IUT-OBE Framework**. The university wants to release it as open-source software so that others can use, modify, and distribute it. However, IUT also wants to ensure that if anyone modifies the framework or builds another software system using it, the resulting software must also remain open source and its source code must be disclosed when distributed. Based on the given scenario, answer the following questions: 3 + 4 (CO1) (PO1)

   i. What is an Open Source Software (OSS) license?
   ii. Which type of licensing technique should be followed when designing the IUT Framework license to enforce this requirement? Explain the concept.

6. In a software company, two developers, Alice and Bob, are working on the same file using a version control system. The central repository initially contains the sentence: "**Tom the cat.**" Both developers check out the file to their local machines. Alice updates the sentence to "**Tom the British cat,**" while Bob changes it to "**Tom the Bangladeshi cat.**"

a) Now both developers try to commit their changes to the central repository. Will there be any problem? If yes, explain what problem may occur. 4 (CO1) (PO1)

b) Assume the system follows the Copy–Modify–Merge model and Alice commits her change first. If Bob wants to commit his changes next, what must be done before the commit? 5 (CO1) (PO1)

c) In the given scenario, the Lock–Modify–Unlock model may work. However, it may not be suitable for large-scale software projects involving many developers. Do you agree with this statement? Explain briefly. 6 (CO1) (PO1)

d) The Lock–Modify–Unlock model is generally harmful to collaboration, but sometimes locking is necessary. Describe a situation where locking a file would be appropriate. 6 (CO1) (PO1)

e) Modern version control systems such as Git usually use the distributed version control model. Mention two benefits of distributed version control. 4 (CO1) (PO1)